

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 1 to 6 (M2)

AIM :

1. Generating descriptive statistics using summary() or describe() (R).
2. Generating frequency tables using table() or count() (R).
3. Creating cross-tabulations and two-way tables using table() (R).
4. Performing one-sample t-tests using t.test() (R).
5. Performing independent two-sample t-tests using t.test() with grouping (R).
6. Performing paired t-tests using t.test(paired=TRUE) (R).

Input :

```
# Module 2 : Practical 1 to 6
# Dataset : Indian_Climate_Dataset_2024_2025.csv
library(dplyr)
library(psych)
# 0. LOAD DATA
df <- read.csv("Indian_Climate_Dataset_2024_2025.csv",
              na.strings = c("", "NA"),
              stringsAsFactors = FALSE)
# Identify numeric and categorical columns safely
num_cols <- names(df)[sapply(df, is.numeric)]
cat_cols <- names(df)[sapply(df, is.character)]
# Choose primary temperature column (first numeric column)
temp_col <- num_cols[1]
# Create grouping variable
df$Temp_Group <- ifelse(df[[temp_col]] > mean(df[[temp_col]], na.rm = TRUE),
                        "High", "Low")
# 1. Practical: Descriptive statistics
print("1. Descriptive Statistics")
summary(df[[temp_col]])
describe(df[[temp_col]])

# 2. Practical: Frequency tables
print("2. Frequency Table")
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

```
freq_col <- cat_cols[1]
table(df[[freq_col]])
df %>% count(.data[[freq_col]])
```

3. Practical: Cross-tabulation

```
print("3. Cross Tabulation")
if (length(cat_cols) >= 2) {
  table(df[[cat_cols[1]]], df[[cat_cols[2]]])
} else {
  print("Not enough categorical columns for cross-tabulation")
}
```

4. Practical: One-sample t-test

```
print("4. One-Sample t-test")
t_test_one <- t.test(df[[temp_col]], mu = mean(df[[temp_col]], na.rm = TRUE))
print(t_test_one)
```

5. Practical: Independent two-sample t-test

```
print("5. Independent Two-Sample t-test")
t_test_two <- t.test(df[[temp_col]] ~ df$Temp_Group)
print(t_test_two)
```

6. Practical: Paired t-test

```
print("6. Paired t-test")
# Use first two numeric columns for paired test
if (length(num_cols) >= 2) {
  t_test_paired <- t.test(df[[num_cols[1]]],
                        df[[num_cols[2]]],
                        paired = TRUE)
  print(t_test_paired)
} else {
  print("Not enough numeric columns for paired t-test")
}
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

Output :

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
R 4.5.2 · C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 1 to 6 M2/
> # 1. Practical: Descriptive Statistics
> print("--- 1. Descriptive Statistics ---")
[1] "--- 1. Descriptive Statistics ---"
> summary(df[[temp_col]])
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
25.00   30.00   34.90   34.95   40.00   45.00
> describe(df[[temp_col]])
  vars  n mean  sd median trimmed  mad min max range skew kurtosis  se
x1     1 7310 34.95 5.78   34.9   34.95  7.41  25  45   20 0.01  -1.2 0.07

> # 2. Practical: Frequency tables
> print("--- 2. Frequency Table ---")
[1] "--- 2. Frequency Table ---"
> freq_col <- cat_cols[1]
> table(df[[freq_col]])

2024-01-01 2024-01-02 2024-01-03 2024-01-04 2024-01-05 2024-01-06 2024-01-07 2024-01-08
10         10         10         10         10         10         10         10
2024-01-09 2024-01-10 2024-01-11 2024-01-12 2024-01-13 2024-01-14 2024-01-15 2024-01-16
10         10         10         10         10         10         10         10
2024-01-17 2024-01-18 2024-01-19 2024-01-20 2024-01-21 2024-01-22 2024-01-23 2024-01-24
10         10         10         10         10         10         10         10
2024-01-25 2024-01-26 2024-01-27 2024-01-28 2024-01-29 2024-01-30 2024-01-31 2024-02-01
10         10         10         10         10         10         10         10
2024-02-02 2024-02-03 2024-02-04 2024-02-05 2024-02-06 2024-02-07 2024-02-08 2024-02-09
10         10         10         10         10         10         10         10
2024-02-10 2024-02-11 2024-02-12 2024-02-13 2024-02-14 2024-02-15 2024-02-16 2024-02-17
10         10         10         10         10         10         10         10
2024-02-18 2024-02-19 2024-02-20 2024-02-21 2024-02-22 2024-02-23 2024-02-24 2024-02-25
10         10         10         10         10         10         10         10
2024-02-26 2024-02-27 2024-02-28 2024-02-29 2024-03-01 2024-03-02 2024-03-03 2024-03-04
10         10         10         10         10         10         10         10
2024-03-05 2024-03-06 2024-03-07 2024-03-08 2024-03-09 2024-03-10 2024-03-11 2024-03-12
10         10         10         10         10         10         10         10
2024-03-13 2024-03-14 2024-03-15 2024-03-16 2024-03-17 2024-03-18 2024-03-19 2024-03-20
10         10         10         10         10         10         10         10
2024-03-21 2024-03-22 2024-03-23 2024-03-24 2024-03-25 2024-03-26 2024-03-27 2024-03-28
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)

Source
R 4.5.2 · C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 1 to 6 M2/

10         10         10         10         10         10         10         10
2025-12-13 2025-12-14 2025-12-15 2025-12-16 2025-12-17 2025-12-18 2025-12-19 2025-12-20
10         10         10         10         10         10         10         10
2025-12-21 2025-12-22 2025-12-23 2025-12-24 2025-12-25 2025-12-26 2025-12-27 2025-12-28
10         10         10         10         10         10         10         10
2025-12-29 2025-12-30 2025-12-31
10         10         10

> df %>% count(.data[[freq_col]])
  date n
1 2024-01-01 10
2 2024-01-02 10
3 2024-01-03 10
4 2024-01-04 10
5 2024-01-05 10
6 2024-01-06 10
7 2024-01-07 10
8 2024-01-08 10
9 2024-01-09 10
10 2024-01-10 10
11 2024-01-11 10
12 2024-01-12 10
13 2024-01-13 10
14 2024-01-14 10
15 2024-01-15 10
16 2024-01-16 10
17 2024-01-17 10
18 2024-01-18 10
19 2024-01-19 10
20 2024-01-20 10
21 2024-01-21 10
22 2024-01-22 10
23 2024-01-23 10
24 2024-01-24 10
25 2024-01-25 10
26 2024-01-26 10
27 2024-01-27 10
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)
Source
R v. 4.5.2 · C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 1 to 6 M2/
500 2025-05-14 10
[ reached 'max' / getOption("max.print") -- omitted 231 rows ]
> # 3. Practical: cross-tabulation
> print("--- 3. cross Tabulation ---")
[1] "--- 3. cross Tabulation ---"
> if (length(cat_cols) >= 2) {
+   table(df[[cat_cols[1]]], df[[cat_cols[2]])
+ } else {
+   print("Not enough categorical columns for cross-tabulation")
+ }

Ahmedabad Bengaluru Bhopal Chennai Delhi Hyderabad Jaipur Kolkata Lucknow
2024-01-01      1      1      1      1      1      1      1      1      1
2024-01-02      1      1      1      1      1      1      1      1      1
2024-01-03      1      1      1      1      1      1      1      1      1
2024-01-04      1      1      1      1      1      1      1      1      1
2024-01-05      1      1      1      1      1      1      1      1      1
2024-01-06      1      1      1      1      1      1      1      1      1
2024-01-07      1      1      1      1      1      1      1      1      1
2024-01-08      1      1      1      1      1      1      1      1      1
2024-01-09      1      1      1      1      1      1      1      1      1
2024-01-10      1      1      1      1      1      1      1      1      1
2024-01-11      1      1      1      1      1      1      1      1      1
2024-01-12      1      1      1      1      1      1      1      1      1
2024-01-13      1      1      1      1      1      1      1      1      1
2024-01-14      1      1      1      1      1      1      1      1      1
2024-01-15      1      1      1      1      1      1      1      1      1
2024-01-16      1      1      1      1      1      1      1      1      1
2024-01-17      1      1      1      1      1      1      1      1      1
2024-01-18      1      1      1      1      1      1      1      1      1
2024-01-19      1      1      1      1      1      1      1      1      1
2024-01-20      1      1      1      1      1      1      1      1      1
2024-01-21      1      1      1      1      1      1      1      1      1
2024-01-22      1      1      1      1      1      1      1      1      1
2024-01-23      1      1      1      1      1      1      1      1      1
2024-01-24      1      1      1      1      1      1      1      1      1
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)
Source
R v. 4.5.2 · C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 1 to 6 M2/
> # 4. Practical: One-sample t-test
> print("--- 4. One-Sample t-test ---")
[1] "--- 4. One-Sample t-test ---"
> t_test_one <- t.test(df[[temp_col]], mu = mean(df[[temp_col]]), na.rm = TRUE)
> print(t_test_one)

One Sample t-test

data: df[[temp_col]]
t = 0, df = 7309, p-value = 1
alternative hypothesis: true mean is not equal to 34.95216
95 percent confidence interval:
 34.81961 35.08472
sample estimates:
mean of x
 34.95216

> # 5. Practical: Independent two-sample t-test
> print("--- 5. Independent Two-Sample t-test ---")
[1] "--- 5. Independent Two-Sample t-test ---"
> t_test_two <- t.test(df[[temp_col]] ~ df$Temp_Group)
> print(t_test_two)

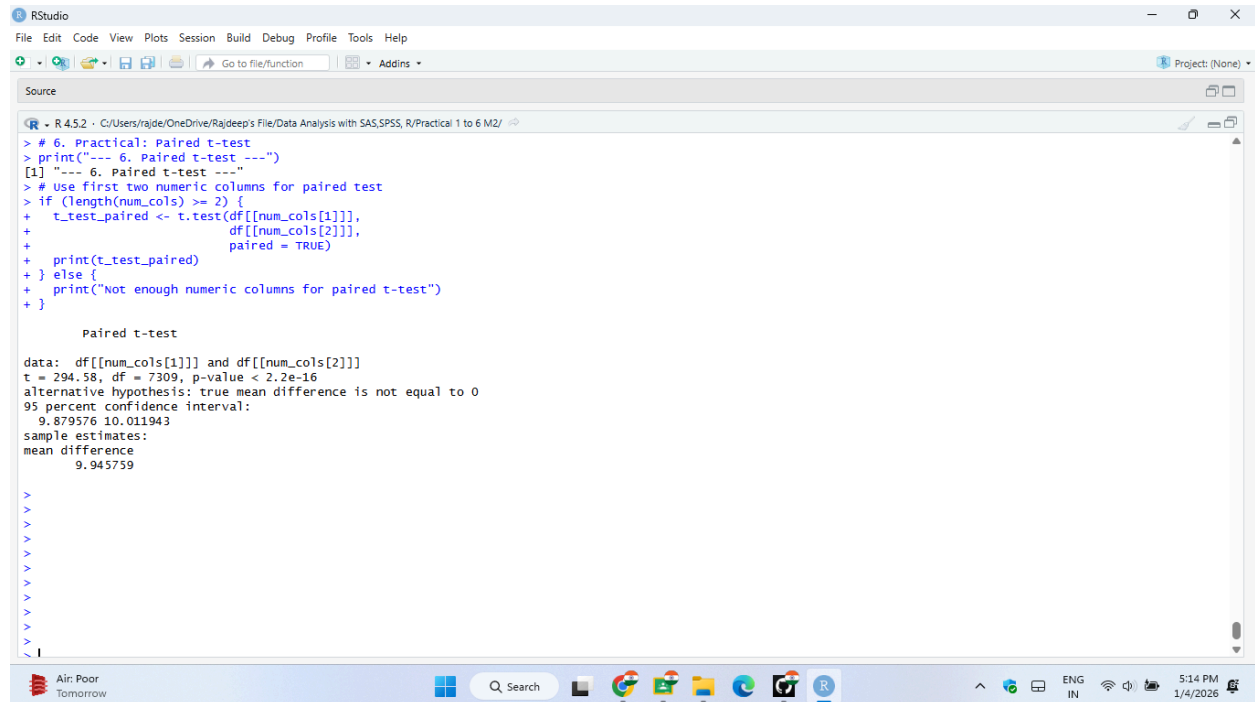
Welch Two Sample t-test

data: df[[temp_col]] by df$Temp_Group
t = 148.51, df = 7303.8, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 9.888299 10.152831
sample estimates:
mean in group High mean in group Low
 39.97204 29.95147

> # 6. Practical: Paired t-test
> print("--- 6. Paired t-test ---")
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical



The screenshot shows the RStudio interface with a script editor containing an R script for a paired t-test. The script defines a function that takes a data frame and returns the results of a paired t-test. The output in the console shows the results of the test, including the t-value, degrees of freedom, p-value, and confidence interval.

```
> # 6. Practical: Paired t-test
> print("--- 6. Paired t-test ---")
[1] "--- 6. Paired t-test ---"
> # Use first two numeric columns for paired test
> if (length(num_cols) >= 2) {
+   t_test_paired <- t.test(df[[num_cols[1]]],
+                           df[[num_cols[2]]],
+                           paired = TRUE)
+   print(t_test_paired)
+ } else {
+   print("Not enough numeric columns for paired t-test")
+ }
```

Paired t-test

data: df[[num_cols[1]]] and df[[num_cols[2]]]
t = 294.58, df = 7309, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
9.879576 10.011943
sample estimates:
mean difference
9.945759